

Topic P4

Electricity and magnetism

Concept guide · Worked examples · Cambridge-style practice · Fully explained answers

What this guide teaches

A complete explanation of the syllabus concepts, common misconceptions, worked applications and exam-style practice for this topic.

Syllabus map

P4.1	Simple phenomena of magnetism
P4.2	Electrical quantities
P4.3	Electric circuits
P4.4	Electrical safety
P4.5	Electromagnetic effects

How to study: retrieve definitions and equations, cover the worked examples, then complete the practice paper without using the notes.

1. Charge, current and potential difference

Electric charge is positive or negative. Current is the rate of flow of charge; conventional current is from positive to negative outside a source. Potential difference is energy transferred per unit charge, while e.m.f. is energy supplied per unit charge by a source.

$$I = Q/t \quad V = E/Q$$

Worked example

A current of 0.40 A flows for 150 s. Charge = $It = 60 \text{ C}$.

Examiner insight

Distinguish e.m.f. of a source from p.d. across a component.

2. Resistance and current-voltage behaviour

Resistance opposes current. For an ohmic conductor at constant temperature, current is proportional to voltage. A filament lamp heats as current rises, increasing resistance; a diode conducts mainly in one direction.

$$R = V/I$$

Worked example

A 12 V component carries 0.30 A. $R = 12/0.30 = 40 \Omega$.

Examiner insight

Do not assume every component has constant resistance.

3. Series and parallel circuits

In series, current is the same everywhere, p.d.s add and resistances add. In parallel, p.d. is the same across branches and branch currents add at a junction. Adding a parallel branch reduces total resistance and increases source current.

Examiner insight

Use conservation of charge to explain current at a junction.

4. Electrical energy, power and safety

Electrical power is the rate of energy transfer. Fuses and circuit breakers interrupt excessive current; earth wires provide a low-resistance safety path; double insulation removes the need for an earth connection.

$$P = IV \quad E = IVt \quad P = I^2R \quad P = V^2/R$$

Worked example

A 230 V kettle draws 8.0 A. Power = $230 \times 8.0 = 1840$ W.

Examiner insight

A fuse rating should be just above the normal operating current.

5. Magnetic fields and electromagnets

Field lines run from north to south outside a magnet and never cross. A current produces a magnetic field; a solenoid field resembles a bar magnet. Electromagnet strength increases with current, turns per length and an iron core.

Examiner insight

A magnetic field is a region in which a magnetic pole experiences a force.

6. Motor effect and electromagnetic induction

A current-carrying conductor in a magnetic field experiences a force perpendicular to current and field. A coil turns because forces form a couple. Induction occurs when magnetic flux linkage changes; faster change, stronger field or more turns gives larger induced e.m.f. The induced effect opposes the change.

transformer ratio: $V_p/V_s = N_p/N_s$

Worked example

A 240 V transformer has 1200 primary turns and 60 secondary turns. $V_s = 240 \times 60/1200 = 12$ V.

Examiner insight

A transformer requires alternating current because the magnetic field must change.

Exam technique

High-value habits

Use precise definitions, show every calculation step, include units, apply concepts to unfamiliar situations, and answer the command word directly.

Retrieval checklist

- Recall all definitions and equations without notes.
- Explain one unfamiliar application using a linked chain of reasoning.
- Identify the first error in each incorrect response.
- Complete one timed set and mark it only after finishing.

Practice Paper

Topic P4

Electricity and magnetism

Instructions

Answer all questions. Select one option for multiple choice. Show working in open spaces and write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes.

Section A · Multiple choice

Choose the best answer for each question.

1. Which equation defines current? [1]
 - A $I=Q/t$
 - B $I=Qt$
 - C $I=VQ$
 - D $I=E/t$

2. A 6.0 V resistor carries 0.50 A. Its resistance is [1]
 - A 3.0 Ω
 - B 12 Ω
 - C 30 Ω
 - D 0.083 Ω

3. What is the current at every point in a series circuit? [1]
 - A always zero
 - B the same
 - C different at each resistor
 - D equal to voltage

4. Which quantity is the same across parallel branches? [1]
 - A current
 - B charge
 - C potential difference
 - D resistance

5. A 230 V appliance draws 4.0 A. Its power is [1]
 - A 57.5 W
 - B 226 W
 - C 234 W
 - D 920 W

6. Which wire is a low-resistance safety path to ground? [1]
 - A live
 - B neutral
 - C earth
 - D filament

7. Outside a bar magnet, field lines travel from [1]
 - A south to north
 - B north to south
 - C east to west
 - D both poles outward

8. How can a solenoid field be strengthened? [1]
- A decrease current
 - B remove iron core
 - C increase turns per length
 - D use direct current of zero value
9. When is an e.m.f. induced in a coil? [1]
- A magnetic flux linkage changes
 - B the coil is stationary in a constant field
 - C current is zero everywhere
 - D the magnet is absent
10. A step-down transformer has [1]
- A more secondary than primary turns
 - B fewer secondary than primary turns
 - C equal turns always
 - D a d.c. output only

Section B · Structured questions

11 A current of 0.75 A flows for 4.0 minutes.

(a) Calculate the charge transferred. [2]

(b) State what conventional current represents. [2]

(c) A component transfers 540 J when 180 C passes through it. Calculate the p.d. [2]

(d) Distinguish between p.d. and e.m.f. [2]

[Total: 8]

12 Two resistors, 4.0 Ω and 8.0 Ω are connected in series to a 12 V supply.

(a) Calculate total resistance. [1]

(b) Calculate the circuit current. [2]

(c) Calculate the p.d. across the 8.0 Ω resistor. [2]

(d) Explain what happens to total resistance if the resistors are connected in parallel. [3]

[Total: 8]

13 A 230 V heater has a power rating of 2.0 kW.

(a) Calculate its normal operating current. [2]

(b) Choose a suitable fuse rating from 3 A, 10 A and 30 A and explain. [2]

(c) Calculate energy transferred in 5.0 minutes. [2]

(d) State two features that protect a user from electric shock. [2]

[Total: 8]

14 A student makes an electromagnet using a coil around an iron core.

(a) State two ways to increase its strength. [2]

(b) Describe the magnetic field inside a long solenoid. [2]

(c) Explain why the core is iron rather than steel when the magnet must switch off. [2]

(d) State two uses of electromagnets. [2]

[Total: 8]

15 A transformer has 1500 primary turns and 100 secondary turns. The primary voltage is 240 V.

(a) Calculate the secondary voltage. [2]

(b) State whether it is step-up or step-down. [1]

(c) Explain why a transformer does not operate with steady d.c. [2]

(d) State three factors that increase induced e.m.f. in a coil. [3]

[Total: 8]

Worked solutions

Remove or fold back this section while completing the practice paper.

Marking routine

Award marks for correct physics and method, not only the final value. Mark each part separately.

Solutions · Multiple choice

1 A

2 B

3 B

4 C

5 D

6 C

7 B

8 C

9 A

10 B

Solutions · Structured questions

11. 11(a) $Q=It=0.75 \times 240=180$ C [2]. (b) Flow of positive charge [2]. (c) $V=E/Q=540/180=3.0$ V [2]. (d) p.d. is energy transferred per charge; e.m.f. is energy supplied per charge [2].
12. 12(a) $12\ \Omega$ [1]. (b) $I=V/R=12/12=1.0$ A [2]. (c) $V=IR=1.0 \times 8.0=8.0$ V [2]. (d) Total resistance is less than either branch because current has additional paths [3].
13. 13(a) $I=P/V=2000/230=8.7$ A [2]. (b) 10 A, just above operating current [2]. (c) $E=Pt=2000 \times 300=6.0 \times 10^5$ J [2]. (d) Any two valid safety features [1 each].
14. 14(a) Increase current; add turns; use an iron core [any 2]. (b) Strong, nearly uniform, parallel field lines [2]. (c) Iron demagnetises easily; steel retains magnetism [2]. (d) Any two valid uses [2].
15. 15(a) $V_s=240 \times 100/1500=16$ V [2]. (b) Step-down [1]. (c) Steady d.c. gives no changing magnetic flux, so no induced secondary e.m.f. [2]. (d) Faster change, stronger magnet/field, more turns [3].

Self-assessment

Circle the first incorrect step and classify it as knowledge, equation choice, rearrangement, unit conversion, graph/diagram interpretation or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.